

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12416901

Kind Code

B2

Date of Patent

September 16, 2025

Inventor(s)

Samardzija; Miroslav et al.

Electronic device with near-field antenna operating through display

Abstract

An electronic device may have a display. A display cover layer and a transparent inner display member may overlap a display pixel layer. The display pixel layer may have an array of display pixels for displaying images for a user. A touch sensor layer may be interposed between the display pixel layer and the transparent display member. A ferromagnetic shielding layer may be mounted below the display pixel layer. A flexible printed circuit containing coils of metal signal lines that form a near-field communications loop antenna may be interposed between the ferromagnetic shielding layer and the display pixel layer. A non-near-field antenna such as an inverted-F antenna may have a resonating element mounted on an inner surface of the display cover layer. The resonating element may be interposed between the transparent display member and the display cover layer.

Inventors: Samardzija; Miroslav (Mountain View, CA), Wang; Yiren (Cupertino, CA), Ouyang; Yuehui (Sunnyvale, CA), Hakim; Joseph (Jacksonville, OR), Li; Qingxiang (Cupertino, CA), Schlub; Robert W. (Palo Alto, CA), Caballero; Ruben (San Jose, CA), Yong; Siwen (San Francisco, CA), de Jong; Erik G. (San Francisco, CA)

Applicant: Apple Inc. (Cupertino, CA)

Family ID: 1000008820161

Appl. No.: 18/525528

Filed: November 30, 2023

Prior Publication Data

Document Identifier

Publication Date

US 20240097309 A1

Mar. 21, 2024

Related U.S. Application Data

continuation parent-doc US 17527906 20211116 US 11894600 child-doc US 18525528
continuation parent-doc US 16596593 20191008 US 11205832 20211221 child-doc US 17527906
continuation parent-doc US 15090781 20160405 US 9685690 20170620 child-doc US 15625903
continuation parent-doc US 14259861 20140423 US 9356661 20160531 child-doc US 15090781
division parent-doc US 15625903 20170616 US 10461395 20191029 child-doc US 16596593

Publication Classification

Int. Cl.: **G04R60/10** (20130101); **G04G21/04** (20130101); **G06F1/16** (20060101); **H01Q1/22** (20060101); **H01Q1/27** (20060101); **H01Q1/52** (20060101); **H01Q7/00** (20060101); **H01Q9/04** (20060101); **H01Q9/42** (20060101); **H01Q21/28** (20060101); **H04B5/26** (20240101); **H04B5/48** (20240101); **H04B5/70** (20240101); **H04B5/72** (20240101); **H05K1/02** (20060101); **H05K5/03** (20060101); **H10K59/40** (20230101)

U.S. Cl.:

CPC **G04R60/10** (20130101); **G04G21/04** (20130101); **G06F1/163** (20130101); **H01Q1/22** (20130101); **H01Q1/273** (20130101); **H01Q1/52** (20130101); **H01Q1/526** (20130101); **H01Q7/00** (20130101); **H01Q9/0421** (20130101); **H01Q9/42** (20130101); **H01Q21/28** (20130101); **H04B5/26** (20240101); **H04B5/48** (20240101); **H04B5/70** (20240101); **H05K1/0278** (20130101); **H05K5/03** (20130101); **H10K59/40** (20230201); **H04B5/72** (20240101)

Field of Classification Search

CPC: H01Q (1/243); H01Q (7/00); H01Q (1/44); H01Q (1/38); H01Q (9/42); H01Q (1/2225); H01Q (1/2258); H01Q (1/2266); H01Q (1/24); H01Q (1/273); H01Q (1/36); H01Q (1/42); H01Q (1/48); H01Q (13/10); H01Q (13/18); H01Q (21/00); H01Q (5/364); H01Q (7/04); H01Q (9/0421); H01Q (9/0428); H01Q (9/045); G06F (1/1626); G06F (3/041); G06F (3/0412); G06F (3/0445); G06F (3/0446); G06F (1/1637); G06F (1/1643); G06F (1/1698); G06F (2203/04107); G06F (3/0443); H04B (5/0031); H04B (5/0081); H04B (5/0037); H04B (5/02); H04B (1/3827); H04B (15/00); H04B (5/00); H04B (5/0025); H04B (5/0075); H04M (1/0266); H04M (1/72412); H04M (2250/22); H04M (1/0237); H04M (1/0249); H04M (1/0254); H04M (1/0274); H04M (1/0277); H04M (1/11); H04M (1/185); H04M (1/72478)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6825751	12/2003	Kita et al.	N/A	N/A
9626024	12/2016	Tan	N/A	H05K 13/00
2011/0006953	12/2010	Chiang	343/702	H01Q 1/2266
2011/0210825	12/2010	Murakami et al.	N/A	N/A
2011/0222248	12/2010	Sakai et al.	N/A	N/A
2011/0241949	12/2010	Nickel et al.	N/A	N/A
2012/0028679	12/2011	Ozasa	N/A	N/A
2012/0133597	12/2011	Chen	N/A	N/A

2012/0223865	12/2011	Li et al.	N/A	N/A
2013/0044074	12/2012	Park et al.	N/A	N/A
2013/0050032	12/2012	Shiu et al.	N/A	N/A
2013/0052947	12/2012	Kole et al.	N/A	N/A
2013/0078917	12/2012	Cho et al.	N/A	N/A
2013/0082895	12/2012	Shiu	343/866	H01Q 7/00
2013/0140965	12/2012	Franklin et al.	N/A	N/A
2013/0229362	12/2012	Liu et al.	N/A	N/A
2013/0265199	12/2012	Koskiniemi et al.	N/A	N/A
2013/0293424	12/2012	Zhu et al.	N/A	N/A
2013/0318766	12/2012	Kiple et al.	N/A	N/A
2014/0028619	12/2013	Huang et al.	N/A	N/A
2014/0080411	12/2013	Konanur et al.	N/A	N/A
2014/0133074	12/2013	Zahler et al.	N/A	N/A
2014/0170975	12/2013	Laio	N/A	N/A
2014/0213178	12/2013	Wolff	455/41.1	H01Q 1/243
2014/0306288	12/2013	Adachi	257/347	H01L 27/1218
2015/0207913	12/2014	Nakano et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2474872	12/2011	EP	N/A
2337150	12/2011	EP	N/A
2869398	12/2014	EP	N/A
10-2014-0024543	12/2013	KR	N/A
201409500	12/2013	TW	N/A
2009014366	12/2008	WO	N/A
2013149515	12/2012	WO	N/A
2013/190119	12/2012	WO	N/A

Primary Examiner: Sorowar; Golam

Attorney, Agent or Firm: Treyz Law Group, P.C.

Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 17/527,906, filed Nov. 16, 2021, which is a continuation of U.S. patent application Ser. No. 16/596,593, filed Oct. 8, 2019, now U.S. Pat. No. 11,205,832, which is a division of U.S. patent application Ser. No. 15/625,903, filed Jun. 16, 2017, now U.S. Pat. No. 10,461,395, which is a continuation of U.S. patent application Ser. No. 15/090,781, filed Apr. 5, 2016, now U.S. Pat. No. 9,685,690, which is a continuation of U.S. patent application Ser. No. 14/259,861, filed Apr. 23, 2014, now U.S. Pat. No. 9,356,661. The disclosures of these applications are hereby incorporated by reference herein in their entireties.

BACKGROUND

(1) This relates to electronic devices, and more particularly, to antennas for electronic devices with wireless communications circuitry.

(2) Electronic devices such as portable computers and cellular telephones are often provided with wireless communications capabilities. For example, electronic devices may use long-range wireless

communications circuitry such as cellular telephone circuitry to communicate using cellular telephone bands. Electronic devices may use short-range wireless communications circuitry such as wireless local area network communications circuitry to handle communications with nearby equipment. Electronic devices may also be provided with satellite navigation system receivers and other wireless circuitry such as near-field communications circuitry. Near-field communications schemes involve electromagnetically coupled communications over short distances, typically 20 cm or less.

(3) To satisfy consumer demand for small form factor wireless devices, manufacturers are continually striving to implement wireless communications circuitry such as antenna components using compact structures. At the same time, there is a desire for wireless devices to cover multiple communications bands. For example, it may be desirable for a wireless device to cover a near-field communications band while simultaneously covering a non-near-field (far field) band.

(4) Because antennas have the potential to interfere with each other and with components in a wireless device, care must be taken when incorporating antennas into an electronic device. Moreover, care must be taken to ensure that the antennas and wireless circuitry in a device are able to exhibit satisfactory performance over a range of operating frequencies.

(5) It would therefore be desirable to be able to provide improved wireless communications circuitry for wireless electronic devices.

SUMMARY

(6) An electronic device may have a housing in which a display is mounted. The electronic device may be a portable electronic device having a wrist strap for attaching the electronic device to the wrist of a user. The electronic device may have wireless circuitry such as near-field communications circuitry and non-near-field communications circuitry.

(7) A display cover layer and a transparent inner display member may overlap a display pixel layer that has an array of display pixels for displaying images to a user. The display cover layer may have an outer surface and an opposing inner surface. The transparent display member may have a curved or planar outer surface that mates with the inner surface of the display cover layer.

(8) A touch sensor layer may be interposed between the display pixel layer and the transparent display member. A ferromagnetic shielding layer may be mounted below the display pixel layer. A flexible printed circuit containing coils of metal signal lines that form a near-field communications loop antenna may be interposed between the ferromagnetic shielding layer and the display pixel layer.

(9) A non-near-field antenna such as an inverted-F antenna may have a resonating element mounted on an inner surface of the display cover layer. The resonating element may be interposed between the transparent display member and the display cover layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an illustrative electronic device such as a laptop computer with wireless circuitry in accordance with an embodiment.

(2) FIG. 2 is a perspective view of an illustrative electronic device such as a handheld electronic device with wireless circuitry in accordance with an embodiment.

(3) FIG. 3 is a perspective view of an illustrative electronic device such as a tablet computer with wireless circuitry in accordance with an embodiment.

(4) FIG. 4 is a perspective view of an illustrative electronic device such as a display for a computer or television with wireless circuitry in accordance with an embodiment.

(5) FIG. 5A is a perspective view of an illustrative electronic device such as a wrist-watch device with wireless circuitry in accordance with an embodiment.

- (6) FIG. 5B is a perspective view of an illustrative electronic device such as a pendant or pin device with wireless circuitry in accordance with an embodiment.
- (7) FIG. 6 is a schematic diagram of an illustrative electronic device with wireless communications circuitry in accordance with an embodiment.
- (8) FIG. 7 is a diagram of a system in which antenna structures in an electronic device are being used to wirelessly communicate with external electrical equipment using near-field communications and non-near-field communications in accordance with an embodiment.
- (9) FIG. 8 is a diagram showing how an electronic device may have a near-field antenna and a non-near-field antenna in accordance with an embodiment.
- (10) FIG. 9 is a diagram of an illustrative non-near-field antenna such as an inverted-F antenna in accordance with an embodiment.
- (11) FIG. 10 is a diagram of an illustrative near-field antenna such as a near-field communications loop antenna in accordance with an embodiment.
- (12) FIG. 11 is a diagram of an illustrative wrist-watch device with an antenna formed adjacent to a display in accordance with an embodiment.
- (13) FIG. 12 is a perspective view of an illustrative wrist-watch device with a slot antenna in accordance with an embodiment.
- (14) FIG. 13 is a cross-sectional side view of an illustrative electronic device with a near-field communications loop antenna wirelessly communicating with an external near-field communications loop antenna using electromagnetic near-field communication signals passing through a display that overlaps the near-field communications loop antenna in accordance with an embodiment.
- (15) FIG. 14 is a cross-sectional side view of an illustrative electronic device such as a wrist-watch device having a near-field communications antenna operating through a display mounted on the curved underside of a display cover layer and having a non-near-field communications antenna mounted under the display cover layer adjacent to the display in accordance with an embodiment.
- (16) FIG. 15 is a cross-sectional side view of a portion of an electronic device such as a wrist-watch device having an antenna mounted under a display cover layer using a plastic carrier in accordance with an embodiment.
- (17) FIG. 16 is a cross-sectional side view of a portion of an electronic device such as a wrist-watch device having a near-field communications antenna operating through a display mounted on the underside of a clear inner display layer with a convex surface that is located under a curved display cover layer and having a non-near-field communications antenna mounted under the display cover layer adjacent to the display in accordance with an embodiment.
- (18) FIG. 17 is a cross-sectional side view of a portion of an electronic device such as a wrist-watch device having a near-field communications antenna operating through a display mounted on the underside of a planar clear display layer that is located under a display cover layer with a convex surface and having a non-near-field communications antenna mounted between the planar display layer and the display cover layer in accordance with an embodiment.

DETAILED DESCRIPTION

(19) Electronic devices may be provided with wireless circuitry. The wireless circuitry may include near-field communications circuitry. For example, a near-field communications transmitter-receiver (“transceiver”) may use a near-field communications antenna to transmit and receive near-field electromagnetic signals at a frequency such as 13.56 MHz. Near-field communications schemes involve near-field electromagnetic coupling between near-field antennas that are separated by a relatively small distance (e.g., 20 cm or less). The near-field communications antennas may be loop antennas. The wireless circuitry may also include cellular network transceiver circuitry, wireless local area network transceiver circuitry, satellite navigation system circuitry, or other non-near-field communications circuitry. The non-near-field communications circuitry can use an antenna to handle radio-frequency signals at frequencies of 700 MHz to 2700 MHz (e.g., 2.4 GHz), 5 GHz, or

other suitable frequencies.

(20) The wireless communications circuitry in an electronic device may be used to support wireless communications in multiple wireless communications bands. The wireless communications circuitry may include antenna structures that include loop antennas, inverted-F antennas, strip antennas, planar inverted-F antennas, slot antennas, hybrid antennas that include antenna structures of more than one type, or other suitable antennas.

(21) Antenna structures may, if desired, be formed from conductive electronic device structures such as conductive electronic device housing structures, stamped metal foil, wires, metal portions of electronic components, and conductive traces such as metal traces on dielectric substrates. The dielectric substrates on which the conductive traces are formed may be printed circuit substrates (e.g., rigid printed circuit boards formed from fiberglass-filled epoxy or other rigid printed circuit board material and/or flexible printed circuits formed from flexible sheets of polyimide or other flexible polymer layers), plastic carriers, glass substrates, ceramic substrates, or other dielectric support structures. If desired, flexible printed circuits containing antenna structures and other antennas may be mounted to structures in an electronic device using adhesive.

(22) Illustrative electronic devices that may be provided with wireless communications circuitry are shown in FIGS. 1, 2, 3, 4, and 5.

(23) Electronic device **10** of FIG. 1 has the shape of a laptop computer and has upper housing **12A** and lower housing **12B** with components such as keyboard **16** and touchpad **18**. Device **10** has hinge structures **20** (sometimes referred to as a clutch barrel) to allow upper housing **12A** to rotate in directions **22** about rotational axis **24** relative to lower housing **12B**. Display **14** is mounted in housing **12A**. Upper housing **12A**, which may sometimes be referred to as a display housing or lid, is placed in a closed position by rotating upper housing **12A** towards lower housing **12B** about rotational axis **24**.

(24) FIG. 2 shows an illustrative configuration for electronic device **10** based on a handheld device such as a cellular telephone, music player, gaming device, navigation unit, or other compact device. In this type of configuration for device **10**, device **10** has opposing front and rear surfaces. Display **14** is mounted on a front face of device **10**. Display **14** may have an exterior layer that includes openings for components such as button **26** and speaker port **21**.

(25) In the example of FIG. 3, electronic device **10** is a tablet computer. In electronic device **10** of FIG. 3, device **10** has opposing planar front and rear surfaces. Display **14** is mounted on the front surface of device **10**. As shown in FIG. 3, display **14** may have an opening to accommodate button **26**.

(26) FIG. 4 shows an illustrative configuration for electronic device **10** in which device **10** is a computer display, a computer that has an integrated computer display, or a television. Display **14** is mounted on a front face of device **10**. With this type of arrangement, housing **12** for device **10** may be mounted on a wall or may have an optional structure such as support stand **23** to support device **10** on a flat surface such as a table top or desk.

(27) FIG. 5A shows an illustrative configuration for electronic device **10** in which device **10** is a wrist-watch device. Display **14** may be mounted on the front face of device **10**. Buttons **27** or other user input-output components may be mounted on the edges of device housing **12**. Device **10** may have a strap attached to housing **12** such as wrist strap **25** so that device **10** may be attached to the wrist of a user. If desired, devices such as device **10** may be worn as pendant type devices (e.g., wrist strap **25** may serve as a neck cord as shown in FIG. 5B). FIG. 5B also shows how a pin such as spring clip structure **29** may be provided on device housing **12** to allow device **10** to be pinned to the clothing of a user. If desired, strap **25** may be detached.

(28) Electronic device **10** may, in general, be a portable electronic device or other suitable electronic device. For example, electronic device **10** may be a laptop computer, a tablet computer, a somewhat smaller device such as a wrist-watch device, pendant device, headphone device, earpiece device, or other wearable or miniature device, a cellular telephone, or a media player. Device **10**

may also be a television, a set-top box, a desktop computer, a computer monitor into which a computer has been integrated, a television, a computer monitor, or other suitable electronic equipment. The configurations for device **10** that are shown in FIGS. **1**, **2**, **3**, **4**, and **5** are merely illustrative.

(29) Display **14** in devices such as devices **10** of FIGS. **1**, **2**, **3**, **4**, and **5** may be a liquid crystal display, an organic light-emitting diode display, a plasma display, an electrophoretic display, an electrowetting display, a display using other types of display technology, or a display that includes display structures formed using more than one of these display technologies. If desired, a touch sensor may be incorporated into display **14**. For example, a layer in display **14** that contains display pixel structures or a separate substrate may be provided with an array of transparent conductive electrodes such as indium tin oxide electrodes. The array of transparent electrodes may be used to gather capacitive touch sensor measurements (i.e., the array of electrodes may be used to form a capacitive touch sensor for display **14**).

(30) Electronic devices such as electronic devices **10** of FIGS. **1**, **2**, **3**, **4**, and **5** may have antenna structures for handling near-field communications (e.g., communications in a near-field communications band such as a 13.56 MHz band or other near-field communications band) and non-near-field communications (sometimes referred to as far field communications) such as cellular telephone communications, wireless local area network communications, and satellite navigation system communications. Near-field communications typically involve communication distances of less than about 20 cm and involve magnetic (electromagnetic) near-field coupling between near-field antennas such as loop antennas. Far field communications typically involved communication distances of multiple meters or miles.

(31) Electronic devices such as devices **10** of FIGS. **1**, **2**, **3**, **4**, and **5** may include a housing such as housing **12**. Housing **12**, which may sometimes be referred to as a case, may be formed of plastic, glass, ceramics, fiber composites, metal (e.g., stainless steel, aluminum, etc.), other suitable materials, or a combination of these materials. In some situations, parts of housing **12** may be formed from dielectric or other low-conductivity material. In other situations, housing **12** or at least some of the structures that make up housing **12** may be formed from metal elements.

(32) A schematic diagram of an illustrative configuration that may be used for an electronic device such as device **10** of FIGS. **1**, **2**, **3**, **4**, and **5** is shown in FIG. **6**. As shown in FIG. **6**, electronic device **10** may include control circuitry such as storage and processing circuitry **28**. Storage and processing circuitry **28** may include storage such as hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in storage and processing circuitry **28** may be used to control the operation of device **10**. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio codec chips, application specific integrated circuits, etc.

(33) Storage and processing circuitry **28** may be used to run software on device **10**, such as internet browsing applications, voice-over-internet-protocol (VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. To support interactions with external equipment, storage and processing circuitry **28** may be used in implementing communications protocols. Communications protocols that may be implemented using storage and processing circuitry **28** include internet protocols, wireless local area network protocols (e.g., IEEE 802.11 protocols—sometimes referred to as WiFi®), protocols for other short-range wireless communications links such as the Bluetooth® protocol, cellular telephone protocols, near-field communications protocols, etc.

(34) Circuitry **28** may be configured to implement control algorithms that control the use of antennas in device **10**. For example, circuitry **28** may perform signal quality monitoring operations, sensor monitoring operations, and other data gathering operations and may, in response to the

gathered data and information on which communications bands are to be used in device **10**, control which antenna structures within device **10** are being used to receive and process data and/or may adjust one or more switches, tunable elements, or other adjustable circuits in device **10** to adjust antenna performance. As an example, circuitry **28** may control which of two or more antennas is being used to receive or transmit near-field or non-near-field wireless signals, which antenna is being used to handle incoming radio-frequency signals, may control which of two or more antennas is being used to transmit radio-frequency signals, may control the process of routing incoming data streams over two or more antennas in device **10** in parallel, may tune an antenna to cover a desired communications band, may perform time-division multiplexing operations to share antenna structures between near-field and non-near-field communications circuitry, to share a non-near-field communications transceiver between multiple non-near-field antennas, to share a near-field communications transceiver between multiple near-field antennas, etc.

(35) In performing these control operations, circuitry **28** may open and close switches (e.g., switches associated with one or more multiplexers or other switching circuitry), may turn on and off receivers and transmitters, may adjust impedance matching circuits, may configure switches in front-end-module (FEM) radio-frequency circuits that are interposed between radio-frequency transceiver circuitry and antenna structures (e.g., filtering and switching circuits used for impedance matching and signal routing), may adjust switches, tunable circuits, and other adjustable circuit elements that are formed as part of an antenna or that are coupled to an antenna or a signal path associated with an antenna, and may otherwise control and adjust the components of device **10**.

(36) Input-output circuitry **30** may be used to allow data to be supplied to device **10** and to allow data to be provided from device **10** to external devices. Input-output circuitry **30** may include input-output devices **32**. Input-output devices **32** may include touch screens, buttons, joysticks, click wheels, scrolling wheels, touch pads, key pads, keyboards, microphones, speakers, tone generators, vibrators, cameras, sensors, light-emitting diodes and other status indicators, data ports, etc. A user can control the operation of device **10** by supplying commands through input-output devices **32** and may receive status information and other output from device **10** using the output resources of input-output devices **32**. The sensors in input-output devices **32** may gather information about the operating environment of device **10** and/or may gather user input. The sensors in devices **32** may include sensors such as a touch sensor, an accelerometer, a compass, a proximity sensor, an ambient light sensor, and other sensors. Sensor data may be used in controlling antenna operation (e.g., in switching between antennas, tuning antennas, etc.).

(37) Wireless communications circuitry **34** may include radio-frequency (RF) transceiver circuitry formed from one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers, passive RF components, one or more antennas, and other circuitry for handling RF wireless signals. Wireless signals can also be sent using light (e.g., using infrared communications).

(38) Wireless communications circuitry **34** may include satellite navigation system receiver circuitry **35** such as Global Positioning System (GPS) receiver circuitry (e.g., circuitry for receiving satellite positioning signals at 1575 MHz) or may include satellite navigation system receiver circuitry associated with other satellite navigation systems.

(39) Wireless local area network transceiver circuitry **36** in wireless communications circuitry **34** may handle 2.4 GHz and 5 GHz bands for WiFi® (IEEE 802.11) communications and may handle the 2.4 GHz Bluetooth® communications band.

(40) Circuitry **34** may use cellular telephone transceiver circuitry **38** for handling wireless communications in cellular telephone bands such as bands in frequency ranges of about 700 MHz to about 2700 MHz or bands at higher or lower frequencies.

(41) Wireless communications circuitry **34** may include near-field communications circuitry **42**. Near-field communications circuitry **42** may handle near-field communications at frequencies such as the near-field communications frequency of 13.56 MHz or other near-field communications

frequencies of interest.

(42) Circuitry **44** such as satellite navigation system receiver circuitry **35**, wireless local area network transceiver circuitry **36**, and cellular telephone transceiver circuitry **38** that does not involve near-field communications may sometimes be referred to as non-near-field communications circuitry or far field communications circuitry.

(43) If desired, communications circuitry **34** may include circuitry for other short-range and long-range wireless links. For example, wireless communications circuitry **34** may include wireless circuitry for receiving radio and television signals, paging circuits, etc. In near-field communications, wireless signals are typically conveyed over distances of less than 20 cm. In WiFi® and Bluetooth® links and other short-range wireless links, wireless signals are typically used to convey data over 20 cm (e.g., 20 cm to hundreds of meters). In cellular telephone links and other long-range links, wireless signals are typically used to convey data over hundreds of meters or thousands of meters.

(44) Wireless communications circuitry **34** may include antennas **40**. Antennas **40** may include near-field and non-near-field antennas. Antennas **40** may be formed using any suitable types of antenna. For example, antennas **40** may include antennas with resonating elements that are formed from loop antenna structures, patch antenna structures, inverted-F antenna structures, closed and open slot antenna structures, planar inverted-F antenna structures, helical antenna structures, strip antennas, monopoles, dipoles, hybrids of these designs, etc. Different types of antennas may be used for different bands and combinations of bands. For example, one type of antenna may be used in forming a local wireless link antenna and another type of antenna may be used in forming a remote wireless link or one type of antenna may be used in forming a non-near-field antenna while another type of antenna may be used in forming a near-field antenna.

(45) FIG. 7 is a schematic diagram showing how antennas **40** in device **10** may be used by near-field communications circuitry **42** and non-near-field communications circuitry **44**. As shown in FIG. 7, electronic device **10** includes control circuitry **28** and input-output devices **32**. Control circuitry **28** may use input-output devices **32** to provide output to a user and to receive input. Control circuitry **28** may use wireless transceiver circuitry **50** and antennas **40** to communicate with external equipment over one or more wireless communications bands including bands for non-near-field communications and near-field communications. Antennas **40** may include one or more near-field-communications antennas and/or one or more non-near-field communications antennas.

(46) Near-field communications circuitry **42** and non-near-field communications circuitry **44** may be coupled to antennas **40**. Near-field communications circuitry **42** (e.g., a near-field communications transceiver) may use a near-field communications antenna to communicate with external near-field communications equipment **58** over near-field communications link **64**. Non-near-field communications circuitry such as radio-frequency transceiver circuitry **44** may use a non-near-field antenna to communicate with a cellular telephone network, a wireless local area network such as an external network of computing equipment, a cellular telephone, a music player, a laptop computer, a desktop computer, a computer integrated into a display, a wireless television, or other far field (non-near-field) wireless network equipment **54** over non-near-field communications wireless link **56** (e.g., a Bluetooth® link, a WiFi® link, a cellular telephone link, etc.).

(47) External equipment such as external equipment **58** may communicate with near-field communications circuitry **42** via magnetic induction. Equipment **58** may include a loop antenna such as loop antenna **62** that is controlled by control circuitry **60**. Loop antenna **62** and one or more loop antennas in antennas **40** may be electromagnetically coupled to support near-field wireless communications when loop antenna **62** and the loop antenna(s) in structures **40** are within an appropriately close distance of each other such as 20 cm or less, as indicated by near-field communications signals **64** of FIG. 7.

(48) Device **10** may use near-field communications circuitry **42** and the near-field communications

loop antenna(s) of antennas **40** to communicate with external near-field communications equipment **58** using passive or active communications. In passive communications, device **10** may use near-field communications circuitry **42** and a near-field communications antenna to modulate electromagnetic signals **64** from equipment **58**. In active communications, near-field communications circuitry **42** and a near-field communications antenna may transmit radio-frequency electromagnetic signals **64** to external equipment **58**.

(49) To provide antennas **40** with the ability to cover communications frequencies of interest, antennas **40** may be provided with circuitry such as filter circuitry (e.g., one or more passive filters and/or one or more tunable filter circuits). Discrete components such as capacitors, inductors, and resistors may be incorporated into the filter circuitry. Capacitive structures, inductive structures, and resistive structures may also be formed from patterned metal structures (e.g., part of an antenna).

(50) If desired, antennas **40** may be provided with adjustable circuits such as tunable circuitry **52**. Tunable circuitry **52** may be controlled by control signals from control circuitry **28**. For example, control circuitry **28** may supply control signals to tunable circuitry **52** via control path **66** during operation of device **10** whenever it is desired to tune antennas **40** to cover a desired communications band (e.g., a desired non-near-field communications band). Paths **68** may be used to convey data between control circuitry **28** and transceiver circuitry **50**.

(51) Passive filter circuitry in antennas **40** may help antennas **40** exhibit antenna resonances in communications bands of interest (e.g., passive filter circuitry in antennas **40** may short together different portions of antennas **40** and/or may form open circuits or pathways of other impedances between different portions of antennas **40** to ensure that desired antenna resonances are produced).

(52) Transceiver circuitry **50** may be coupled to antennas **40** by signal paths such as signal paths **70** and **72**. Signal paths **70** and **72** may include transmission lines, portions of conductive housing structures, ground plane structures, traces on printed circuits, or other conductive paths.

(53) Impedance matching circuitry formed from components such as inductors, resistors, and capacitors may be used in matching the impedance of antennas **40** to the impedance of transmission line structures coupled to antennas **40**. Filter circuitry may also be provided in the transmission line structures and/or antennas **40**. Matching network components may be provided as discrete components (e.g., surface mount technology components) or may be formed from housing structures, printed circuit board structures, traces on plastic supports, etc. Components such as these may also be used in forming passive filter circuitry in antennas **40** and tunable circuitry **52** in antennas **40**.

(54) A transmission line may be coupled between transceiver **44** and antenna feed structures associated with antennas **40**. As an example, antennas **40** may form one or more non-near-field communications antennas such as one or more inverted-F antennas each having an antenna feed with a positive antenna feed terminal and a ground antenna feed terminal. For each non-near-field antenna, a positive transmission line conductor may be coupled to the positive antenna feed terminal and a ground transmission line conductor may be coupled to the ground antenna feed terminal. Each near-field communications antenna may have a pair of antenna feed terminals (e.g., a pair of feed terminals for receiving differential near-field communications signals from near-field communications circuitry **42**). If desired, other types of antenna feed arrangements may be used to couple non-near-field communications transceiver **44** to non-near-field antennas and to couple near-field communications transceiver **42** to near-field antennas.

(55) Tunable circuitry **52** may be formed from one or more tunable circuits such as circuits based on capacitors, resistors, inductors, and switches. Tunable circuitry **52** and filter circuitry in antennas **40** may be implemented using discrete components mounted to a printed circuit such as a rigid printed circuit board (e.g., a printed circuit board formed from glass-filled epoxy) or a flexible printed circuit formed from a sheet of polyimide or a layer of other flexible polymer or may be implemented using circuitry on a plastic carrier, a glass carrier, a ceramic carrier, or other dielectric

substrate. During operation of device **10**, control circuitry **28** may issue commands on path **66** to adjust switches, variable components, and other adjustable circuitry in tunable circuitry **52**, thereby tuning antennas **40**. If desired, tunable circuitry **52** may include one or more inductors. A switch circuit may be used to selectively switch a desired number of the inductors into use. By varying the inductance of tunable circuitry **52** in this way, antennas **40** can be tuned to cover desired communications bands. Tunable circuitry **52** may also include one or more capacitors that are selectively switched into use with a switching circuit to tune antennas **40**. Capacitance adjustments and inductance adjustments may be made using a tunable circuit with adjustable capacitors and inductors and/or separately adjustable capacitor circuits and inductor circuits may be used in tuning antennas **40**. If desired, antenna(s) **40** may include one or more non-tunable (fixed) antennas.

(56) FIG. **8** is a schematic diagram of illustrative wireless circuitry **34** in device **10** in a configuration in which antennas **40** include near-field communications antenna **40N** and non-near-field (far field) communications antenna **40F**.

(57) Non-near-field communications circuitry **44** may wirelessly communicate with external equipment such as equipment **54** of FIG. **7** using non-near-field communications antenna **40F**. Positive antenna feed terminal **84** and ground antenna feed terminal **86** form a non-near-field communications antenna feed that is coupled to non-near-field communications circuitry **44** (e.g., a non-near-field communications transceiver such as a cellular telephone transceiver, wireless local area network transceiver, etc.) by a pair of transmission line conductors in path **70**.

(58) Near-field communications circuitry **42** may wirelessly communicate with external near-field communications equipment **58** using near-field communications antenna **40N**. Near-field communications circuitry **42** (e.g., a near-field communications transceiver operating at 13.56 MHz or other suitable near-field communications frequency) may be coupled to antenna structures **40** using a pair of conductive lines in path **70**, which are coupled to respective near-field antenna feed terminals **94** and **96**. Near-field communications circuitry **42** may have a pair of differential signal terminals (sometimes referred to as +V and -V terminals) for handling differential near-field communications signals. The two differential signal terminals of near-field communications circuitry **42** may be coupled respectively to the two signal lines in path **70**. Near-field communications circuitry **42** may include an amplifier such as amplifier **74**. Amplifier **74** may strengthen near-field communications signals so that near-field communications antenna **40N** may operate satisfactorily, even in the presence of intervening structures between antenna **40N** and loop antenna **62** such as intervening display structures.

(59) Antenna **40F** may be formed using a planar inverted-F antenna, a loop antenna, a monopole, a dipole, a patch antenna, a slot antenna, or any other suitable type of antenna. In the illustrative configuration of FIG. **9**, antenna **40F** has been formed using an inverted-F antenna design.

(60) As shown in FIG. **9**, antenna **40F** may include inverted-F antenna resonating element **76** and a conductive structure such as antenna ground **88**. Antenna resonating element **76** and antenna ground **88** may be formed from metal traces on a flexible printed circuit, metal traces on a rigid printed circuit board, metal traces on other dielectric carriers, portions of an electronic device housing such as a metal midplate structure or internal frame structures, metal housing walls or other portions of housing **12**, conductive structures such as metal portions of electrical components in device **10**, or other conductive structures. As an example, inverted-F antenna resonating element **76** may be formed from metal traces on dielectric substrate **92**. Substrate **92** may be a flexible printed circuit having a flexible dielectric substrate, a rigid printed circuit board, a glass or plastic structure, or other dielectric substrate.

(61) Antenna resonating element **76** may include main antenna resonating element arm **78** (e.g., an arm formed from metal traces on a printed circuit or other dielectric substrate). Main antenna resonating element arm **78** may have one or more branches. For example, arm **78** may have a low band arm LB for producing a low communications band resonance and a high band arm HB for producing a high communications band resonance. Arm **78** may be separated from ground plane **88**

by a dielectric-filled opening such as gap **90**. Gap **90** may contain plastic, glass, ceramic, air, or other dielectric materials. Non-near-field communications antenna return path **80** in non-near-field communications antenna **40F** may bridge gap **90**. Non-near-field communications antenna feed path **82** may bridge gap **90** in parallel with return path **80**. Antenna feed terminals such as positive antenna feed terminal **84** and ground antenna feed terminal **86** may form a non-near-field communications antenna feed within antenna feed path **82**. The conductive structures of antenna return path **80** and antenna feed path **82** may be formed from metal traces on printed circuits, metal traces on plastic carriers, conductive housing structures, or other conductive structures in device **10**. (62) Impedance matching circuitry, filter circuitry, and tunable circuitry **52** of FIG. **7** may be interposed in paths that bridge gap **90** such as path **80**, feed path **82**, or one or more parallel tuning paths or may be formed in other portions of antenna resonating element **76** and/or may be incorporated into ground structures such as antenna ground **88**.

(63) An illustrative near-field communications loop antenna configuration that may be used for near-field communications antenna **40N** is shown in FIG. **10**. As shown in FIG. **10**, near-field communications antenna **40N** has near-field communications antenna terminals **94** and **96** and is formed from one or more coils (turns) of metal signal lines such as lines **98**. In the illustrative arrangement of FIG. **10**, the coils of loop antenna **40N** are rectangular. If desired, the coils of loop antenna **40N** may be circular, may be oval, may be triangular, may have a combination of straight and curved sections, or may have other suitable shapes

(64) If desired, display **14** or other electrical components may overlap some or all of antenna **40N**. As shown in FIG. **10**, for example, an electrical component such as a display may have a footprint such as footprint **100** that overlaps the coils of antenna **40N** or may have a footprint such as footprint **102** that substantially overlaps antenna **40N** but is surrounded by coils **98**. In these configurations, coils **98** run along the periphery of the overlapping component (i.e., coils **98** follow the rectangular edge of rectangular component footprint **100** or **102** and are therefore located just inside or outside of the edge of the component). The component that overlaps antenna **40N** may be a rectangular organic light-emitting diode display having a rectangular periphery or may be any other rectangular electrical component. The component that overlaps antenna **40N** may also have a non-rectangular periphery. Other fully and partially overlapping arrangements may be used if desired. The illustrative arrangements of FIG. **10** are merely illustrative.

(65) FIG. **11** shows how housing **12** of electronic device **10** may have a first portion such as portion **12M** that includes display **14** and a second portion such as portion **12P** that includes non-near-field antenna **40F**. Portion **12M** may be formed from metal or other suitable housing materials. Portion **12P** may include an antenna cover structure formed from plastic or other dielectric materials so that antenna signals associated with antenna **40F** may be conveyed through portion **12P**. By using dielectric materials to enclose antenna **40F** and by laterally offsetting antenna **40F** in lateral dimension X to the side of display **14**, antenna **40F** can operate satisfactorily (e.g., antenna **40F** can operate without interference from conductive structures in display **14** and/or housing **12M**). If desired, housing **12M** may serve as an antenna ground for antenna **40F**.

(66) FIG. **12** shows how antenna **40F** may be a slot antenna. In the example of FIG. **12**, display **14** has an active area AA. Peripheral border areas in display **14** may be inactive (i.e., free of light-emitting display pixels and, if desired, free of touch sensor structures). As shown in FIG. **12**, antenna **40F** may be formed from a slot in inactive area IA such as slot **104**. Slot **104** may be fed by an antenna feed formed from antenna feed terminals **84** and **86** on opposing sides of slot **104**. Slot **104** may extend entirely through device **10** and housing **12** in dimension Z.

(67) If desired, near-field communications antenna **40N** may operate through an electronic component such as display **14**. This type of arrangement may be used in device **10** of FIG. **11**, device **10** of FIG. **12**, and/or other configurations for device **10**.

(68) As shown in FIG. **13**, display **14** may overlap near-field communications antenna **40N**. Display **14** may include an array of light-emitting display pixels **110** such as light-emitting diode

display pixels or other types of display pixel structure. Display **14** may also include conductive structures **102** (e.g., thin-film transistors, touch sensor electrodes, metal signal line traces, etc.). In configurations in which the conductive structures of display **14** are relatively sparse and/or display **14** is not otherwise sufficiently conductive to completely shield near-field communications antenna **40N**, at least some electromagnetic near-field communications signals **64** can pass through display **14** while being conveyed between coils **98** in near-field communications loop antenna **40N** of electronic device **10** and electromagnetically coupled coils **106** in near-field communications loop antenna **62** of external equipment **58**. Configurations of the type shown in FIG. **13** in which electromagnetic near-field communications signals pass through display **14** during near-field communications may allow a near-field communications antenna to be mounted in a device housing where space is limited.

(69) FIG. **14** is a cross-sectional side view of electronic device **10** of FIG. **5A** or FIG. **5B** in an illustrative configuration in which device **10** has a curved cover layer such as cover layer **120** with a curved (convex) outer surface **136** and an opposing curved (concave) inner surface **138**. Display cover layer **120** may be formed from glass, plastic, fused silica, sapphire, or other transparent material.

(70) Display pixel layer **124** may contain an array of display pixels for displaying images to a user such as viewer **140** who is observing display **14** in direction **142**. The display pixels of layer **124** may be based on organic-light emitting diodes or other display pixel structures. Touch sensor **122** (e.g., a touch sensor based on a two-dimensional pattern of transparent capacitive touch sensor electrodes) may be interposed between display layer **124** and inner surface **138** of display cover layer **120**. Substrate **126** may be a flexible printed circuit substrate or other dielectric substrate. Near-field communications loop antenna **40N** may be formed from coils of metal signal lines **98** in substrate **126**. Components **132** such as integrated circuits and other electrical components may be mounted on one or more printed circuits such as printed circuit **130** in the interior of housing **12**. To prevent electromagnetic near-field communications signals from antenna **40N** from creating eddy currents in metal portions of components **132** that could interfere with the operation of antenna **40N**, ferromagnetic shielding layer **128** may be mounted under near-field communications antenna **40N** (i.e., layer **128** may be formed under substrate **126** and coils **98** of near-field communications loop antenna **40N**). Layers such as layers **122**, **124**, **126**, and **128** may be attached to display cover layer **120** and/or each other using adhesive. The structures of layers **122**, **124**, **126**, and **128** may be formed on separate substrates or may be deposited and patterned on shared support structures.

(71) Opaque masking material such as black ink **134** may be formed on peripheral portions of inner surface **138** of display cover layer **120** to shield internal components in device **10** such as antenna structures from view from viewer **140**. Opaque masking material **134** may be white, black, gray, red, blue, green, silver, gold, or other suitable colors.

(72) During operation, near-field communications antenna **40N** transmits and/or receives near-field communications electromagnetic signals in dimension Z through display layers in display **14** such as touch sensor **122** and display layer **124**.

(73) Non-near-field communications antenna **40F** may be formed from a flexible printed circuit or other dielectric substrate with patterned metal traces that form resonating element **76**. The antenna feed for non-near-field communications antenna **40F** may include a positive antenna feed terminal such as positive antenna feed terminal **84** on resonating element **76**. The antenna feed may also include a ground antenna feed terminal on housing **12** (see, e.g., terminal **86-1**), a ground antenna feed on a printed circuit in device **10** such as printed circuit **130** (see, e.g., terminal **86-2**), or other suitable antenna ground terminal.

(74) As shown in the cross-sectional side view of an illustrative portion of device **10** in FIG. **15**, a plastic carrier or other support structure such as support structure **144** may be used to support antenna resonating element **76**. Support structure **144** may be formed from a compressible material such as foam or may include a foam structure or other biasing structure to help press antenna

resonating element **76** against inner surface **138** of display cover layer **120** (i.e., against opaque masking material **134**). Antenna traces for antenna resonating element **76** may be formed directly on masking material **134**, may be formed on a printed circuit such as a flexible printed circuit that is pressed upwards in dimension Z against masking material **134**, may be formed as patterned metal traces on carrier **144**, or may otherwise be incorporated into device **10**.

(75) FIG. **16** is a cross-sectional side view of an illustrative configuration for device **10** showing how device **10** and display **14** may include internal transparent structures such as transparent display member **146** between display cover layer **120** and display pixel layer **124**. Member **146** may be formed from a clear layer of dielectric and may have planar or curved surfaces.

(76) As shown in FIG. **16**, display cover layer **120** may have a curved (convex) outer surface such as outer surface **136** and an opposing curved (concave) inner surface such as inner surface **138**. Transparent member **146** may have a curved (convex) outer surface such as surface **148** that matches the shape of inner surface **138**. Transparent display member **146** may also have an opposing planar inner surface such as planar surface **150**. A layer of adhesive interposed between surfaces **138** and **148** may, if desired, be used to attach member **146** to display cover layer **120**. Display cover layer **120** and member **146** may be formed from clear glass, plastic, fused silica, sapphire, or other transparent material. Light reflections in display **14** may be minimized by forming layer **120** and member **146** from the same material or from different materials that share the same index of refraction.

(77) Touch sensor **122** may, if desired, be interposed between display pixel layer **124** and planar surface **150** of member **146**. Near field communications antenna **40N** may be formed from looped conductive lines **98** on flexible printed circuit substrate **126**. Flexible printed circuit **126** may be mounted under display pixel layer **124**. Ferromagnetic shielding layer **128** may be mounted under display pixel layer **124** and under substrate **126** of antenna **40N**. Because lower surface **150** is planar in the configuration of FIG. **16**, touch sensor layer **122**, display pixel layer **124**, near-field antenna flexible printed circuit **126**, and ferromagnetic layer **128** are planar in the configuration of FIG. **16**.

(78) The centermost portions of touch sensor layer **122** and display pixel layer **124** form an active area in display **14**. Inactive border region **122'** of touch sensor layer **122** may contain touch sensor drive circuits but are devoid of touch sensor capacitor electrodes. Inactive border region **124'** of display pixel layer **124** may contain display driver circuitry but is devoid of light-emitting display pixels. Opaque masking material such as black ink **134** may be formed in peripheral portions of curved inner surface **138** of display cover layer **120** to hide border region **122'** of touch sensor **122** from view and to hide border region **124'** of display pixel layer **124** from view.

(79) Antenna resonating element **76** for non-near-field antenna **40F** may be mounted on inner surface **138** of display cover layer **120**. Non-near-field antenna **40F** may be fed using positive antenna feed terminal **84** on antenna resonating element **76** and ground antenna feed terminal **86**. Ground antenna feed terminal **86** may be shorted to housing **12**, metal traces on printed circuit **130**, and/or metal traces on other layers in display **14** (e.g., touch sensor traces in touch sensor portion **122'**, display pixel layer portions such as portion **124'**, etc.). A dielectric-filled gap of dimension D may separate antenna resonating element **76** from nearby ground structures such as touch sensor portion **122'**, display pixel layer portion **124'**, metal housing **12**, etc. The dielectric-filled gap may be occupied by a dielectric such as air, plastic, glass, sapphire, fused silica, or other dielectric.

(80) In the illustrative configuration of FIG. **17**, display cover layer **120** has a curved (convex) outer (upper) surface **136** and an opposing planar inner (lower) surface **138**. Clear display member **146** has a mating planar outer (upper) surface (i.e., a surface that has the same curvature as inner surface **138** of display cover structure **120**). Clear display member **146** also has planar inner (lower) surface **150**. Planar display layers such as touch sensor **122**, display pixel layer **124**, near-field communications antenna flexible printed circuit **126** for near-field communications antenna **40N**, and ferromagnetic shield layer **128** may be mounted on planar inner surface **150**. Antenna

structures for antenna **40F** such as antenna resonating element **76** may be interposed between the outer surface of clear display member **146** and inner surface **138** of display cover structure **120**. A layer of black ink or other opaque masking material on surface **138** may, if desired, be used to hide antenna resonating element **76** and other internal structures in device **10** from view by a user. (81) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device comprising: a housing; a curved cover layer for a display coupled to the housing; a display pixel layer for the display that is overlapped by the curved cover layer at an active area of the display; an antenna resonating element for an antenna that is overlapped by the curved cover layer at an inactive area of the display; and a biasing structure configured to press the antenna resonating element toward the curved cover layer at the inactive area of the display, wherein the antenna resonating element is disposed on a dielectric substrate and the biasing structure is configured to press the antenna resonating element toward the curved cover layer at the inactive area of the display by applying a force to the dielectric substrate in a direction toward the curved cover layer.
2. The electronic device defined in claim 1, wherein the antenna resonating element comprises conductive traces on the dielectric substrate.
3. The electronic device defined in claim 2, wherein the dielectric substrate is a flexible printed circuit substrate.
4. The electronic device defined in claim 1, wherein the antenna comprises a first antenna feed terminal on the antenna resonating element and a second antenna feed terminal coupled to an antenna ground for the antenna.
5. The electronic device defined in claim 4, wherein the antenna resonating element comprises an antenna resonating element arm, and the antenna comprises a return path that couples the antenna resonating element arm to the antenna ground.
6. The electronic device defined in claim 5, wherein the dielectric substrate comprises a flexible printed circuit substrate, and the antenna resonating element arm, the return path, and ground conductive traces are formed on the flexible printed circuit substrate.
7. The electronic device defined in claim 6, wherein the antenna ground comprises a portion of the housing and the ground conductive traces.
8. The electronic device defined in claim 1, wherein the curved cover layer has a curved outer surface and an inner surface opposite the curved outer surface, and the biasing structure is configured to press the antenna resonating element against the inner surface of the curved cover layer.
9. The electronic device defined in claim 8, wherein the inner surface is a curved inner surface.
10. The electronic device defined in claim 8 further comprising: opaque material between the curved outer surface and the antenna resonating element.
11. The electronic device defined in claim 1, wherein the biasing structure comprises compressible material.
12. The electronic device defined in claim 1, wherein the biasing structure comprises a compressible foam structure.
13. The electronic device defined in claim 1 further comprising: a strap attached to the housing.
14. The electronic device defined in claim 1, wherein the biasing structure is mounted to the housing.
15. An electronic device comprising: a curved display cover layer; a display pixel layer that overlaps the curved display cover layer; a flexible printed circuit that includes a substrate and

conductive traces on the substrate, the conductive traces forming an antenna resonating element configured to convey radio-frequency signals through the curved display cover layer; and a biasing structure configured to apply a force toward the curved display cover layer that presses the flexible printed circuit toward the curved display cover layer.

16. The electronic device defined in claim 15 further comprising: a housing, wherein the biasing structure is mounted between the housing and the flexible printed circuit.

17. The electronic device defined in claim 15, wherein the biasing structure comprises foam.

18. The electronic device defined in claim 15, wherein the conductive traces form an antenna ground and a return path that is coupled between the antenna resonating element and the antenna ground.

19. An electronic device comprising: a housing; a curved cover layer for a display coupled to the housing; a display pixel layer for the display that is overlapped by the curved cover layer at an active area of the display; an antenna resonating element for an antenna that is overlapped by the curved cover layer at an inactive area of the display; and a biasing structure configured to apply a force toward the curved cover layer that presses the antenna resonating element toward the curved cover layer at the inactive area of the display.

20. The electronic device defined in claim 19, wherein the antenna resonating element is pressed against an inner surface of the curved cover layer.
